

CARBERRY, MB, Canada

WMO: 711700

Lat: 49.9058N

Lon: 99.3575W

Elev: 384

StdP: 96.80

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-30.3	-36.4	0.1	-32.9	-33.4	0.2	-30.1	12.9	-9.6	11.4	-8.8	3.0	270	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	30.1	20.9	28.2	20.1	26.6	18.9	23.4	27.8	21.9	26.1	20.6	24.7	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.5	25.9	20.5	15.9	24.3	19.1	14.5	23.0	71.8	28.0	66.0	26.3	61.2	24.6	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.9	8.8	DB	-36.9	33.9	3.4	1.8	-39.4	35.2	-41.3	36.2	-43.2	37.2	-45.7	38.5	(4)
(5)			WB	-36.8	25.7	3.4	1.7	-39.3	26.9	-41.3	27.9	-43.2	28.9	-45.7	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-15.8	-13.9	-6.6	3.4	10.5	16.3	18.6	17.7	12.8	4.8	-4.7	-13.0	(6)	
(7)		DBStd	13.80	8.53	7.91	8.05	6.22	4.89	3.78	3.16	3.41	4.52	4.96	6.79	7.93	(7)	
(8)		HDD10.0	3602	801	669	516	209	54	2	0	0	22	174	441	712	(8)	
(9)		HDD18.3	5856	1059	903	774	449	247	82	36	53	172	421	690	970	(9)	
(10)		CDD10.0	896	0	0	1	11	69	191	267	238	107	12	0	0	(10)	
(11)		CDD18.3	108	0	0	0	0	3	21	44	33	7	0	0	0	(11)	
(12)		CDH23.3	1397	0	0	0	4	86	247	469	459	127	6	0	0	(12)	
(13)		CDH26.7	359	0	0	0	0	19	59	113	139	28	1	0	0	(13)	
(14)	Wind	WSAvg	4.1	4.2	4.1	4.1	4.6	4.8	4.2	3.4	3.5	3.9	4.4	4.3	4.0	(14)	
(15)	Precipitation	PrecAvg	527	20	18	26	31	81	92	73	64	39	39	19	22	(15)	
(16)		PrecMax	726	45	55	77	59	169	146	118	152	76	88	65	43	(16)	
(17)		PrecMin	417	6	1	1	4	9	41	16	11	8	3	2	5	(17)	
(18)		PrecStd	87	9	14	20	17	36	31	30	31	17	20	14	11	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	4.1	14.6	23.5	29.2	31.2	31.8	33.2	29.8	23.4	14.2	5.0	(19)	
MCWB			1.5	2.0	8.9	12.7	16.4	22.2	23.6	22.2	18.7	14.6	8.7	2.5	(20)		
2%		DB	1.5	1.7	9.2	19.9	25.7	28.2	29.5	30.2	26.8	18.9	9.7	2.0	(21)		
		MCWB	0.4	0.7	6.0	10.9	14.2	19.1	22.4	21.0	17.5	11.7	5.9	0.6	(22)		
5%		DB	-0.8	0.2	5.5	17.1	22.9	26.1	27.7	27.9	24.3	15.6	6.4	-0.4	(23)		
		MCWB	-1.6	-0.5	3.4	9.4	13.2	18.0	21.3	20.1	16.4	10.3	3.8	-1.3	(24)		
10%		DB	-3.3	-2.0	3.1	13.8	20.0	24.2	26.1	25.9	21.5	12.8	3.9	-2.3	(25)		
		MCWB	-3.8	-2.6	1.8	8.1	11.9	17.2	20.3	18.9	15.1	9.0	2.2	-2.8	(26)		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	2.4	9.6	14.0	18.4	24.3	26.0	24.8	21.6	15.1	9.0	2.8	(27)	
MCDWB			2.8	3.7	13.5	21.3	25.2	29.0	28.8	29.6	26.2	22.3	13.7	4.5	(28)		
2%		WB	0.5	0.9	6.4	12.0	16.2	21.6	23.9	22.5	19.2	12.8	6.0	0.9	(29)		
		MCDWB	1.5	1.7	9.5	18.5	21.8	25.3	27.8	27.7	23.9	16.7	8.9	2.1	(30)		
5%		WB	-1.5	-0.5	3.8	10.0	14.7	19.7	22.4	21.1	17.6	11.0	3.9	-1.0	(31)		
		MCDWB	-0.7	0.3	5.7	15.8	19.9	23.6	26.4	26.0	22.4	15.0	6.1	-0.3	(32)		
10%		WB	-3.7	-2.6	2.1	8.4	13.2	18.1	21.1	19.8	16.0	9.2	2.2	-2.6	(33)		
		MCDWB	-3.1	-1.9	3.4	13.4	18.6	22.5	24.8	24.4	20.5	12.5	3.7	-2.0	(34)		
(35)	Mean Daily Temperature Range			MDBR	10.0	10.8	10.5	11.7	12.9	11.5	12.3	13.8	13.0	10.6	9.1	(35)	
5% DB				MCDBR	10.8	10.2	11.4	17.5	18.0	14.9	14.5	17.0	17.5	16.2	12.8	10.5	(36)
				MCWBR	9.8	8.9	8.3	9.8	8.7	7.5	8.3	8.8	9.1	9.6	9.1	9.3	(37)
				MCDBR	10.8	9.6	11.0	15.9	13.7	11.9	13.2	14.5	14.6	15.2	12.2	10.1	(38)
5% WB				MCWBR	9.9	8.5	8.3	9.3	7.9	7.3	8.4	8.5	8.7	9.4	8.8	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.252	0.262	0.291	0.344	0.360	0.369	0.381	0.379	0.342	0.293	0.264	0.236	(40)	
(41)		taud		2.299	2.291	2.359	2.353	2.368	2.380	2.354	2.356	2.443	2.547	2.474	2.381	(41)	
(42)		Ebn at Noon		806	891	923	901	897	889	871	855	855	853	798	786	(42)	
(43)		Edh at Noon		67	87	99	112	116	115	117	111	92	69	57	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.43	2.61	3.93	5.04	5.55	5.93	6.28	5.18	3.76	2.26	1.25	1.02	(44)	
(45)		RadStd		0.11	0.19	0.30	0.35	0.47	0.40	0.27	0.37	0.28	0.29	0.17	0.11	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.75	-1.09	N/A	N/A	N/A	N/A	(46)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	-0.53	-0.73	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page